

Node operations and security framework

NODE-OPS-1.0 · review of github.com/AzielEliab · 2026-09-06 · Aziel Eliab

Bulletproof here means smallest public surface, named refuses, isolate-then-phoenix wait / re-seal, no auto-heal onto a poisoned ID, public tunnels / sites that die with the pull, and SPLIT THE WIRES. It does not mean an unbreakable host or a hostname that climbs back.

1. Fleet review (2026-09-06)

Public repos already point the right way: FragGate hashed registry and refuse ledger; corpus security headers; runtime thin MCP; qnm-node on 127.0.0.1 with APG, PHOENIX-LOCK wait / re-seal, SPLIT THE WIRES, no account resurrection, no auto-heal.

2. Node classes

Class	Fleet	Public surface allowed
Face	azieleliab, corpus, godlock	Pages, cite, llms, software mirror while the site exists. Pull site / revoke token / drop Worker / kill DNS → the public face dies with the pull. Phoenix does not restore it.
Fabric	qnm-node	Loopback API only. Tick is presence+tip. Payload is receiver-pull. Not on public faces.
Door	fraggate, aziel-runtime	Thin MCP + fraggate_call. GET /v1/mesh never enables.

3. Attack surface law

- Public tunnels and sites DIE WITH THE PULL. A Cloudflare Tunnel lives on token + DNS name + account.
- No auto-heal. No account resurrection. No public hostname restore.
- SPLIT THE WIRES: 1s loop and 777s gate never share a socket. No body on the tick socket. No fan-out push.

4. Phoenix loop (wait / re-seal — not public restore)

live → APG miss or tamper → isolate → memorial → PHOENIX-LOCK wait / re-seal (local, no controller hunt). Stop there. Phoenix is not “bring the .uk / public node back.” Neighbors do not phoenix because a neighbor did.

7. Split the wires

- Fast 0.5–1s tick: presence + tip hash only. Fixed-size. No body / diff / file on that socket.
- Payload on a second plane the receiver PULLS — never sender fan-out push.
- Update is a proof not a timer: cite prev + lockset, fail-closed verify. 777s = dwell after valid cite (not wait-then-accept). Clock desync ≠ yes. Ambiguous tip = isolate not merge.
- Equivocation: same prev, two tips from one node → lock / isolate that peer. No vote-to-reconcile. Quorum cannot outvote a broken hash.
- Emit last locally. Phoenix is local reboot / WAIT for the failed node only. No unsend of unverified body. Public tunnels die with the pull.
- Partition: split brain keeps separate chains, no auto-splice. Rejoin = cite + operator / lockset. Heartbeat loss ≠ poison and ≠ apply last packet.
- The 1s loop and the 777s gate never share a socket.

8. Close tests

- Phoenix does not restore a public hostname. Pulled sites stay down (die with the pull).
- Tick socket carries no body. Fan-out push refused. Cite fail-closed. 1s and 777s sockets stay split.
- Equivocation locks the peer. Neighbors do not phoenix. No auto-splice. Heartbeat loss does not apply.

Public identity: Aziel Eliab only.

If the files hold, the name was never the point.